



EE 240 Power Engineering Laboratory Indian Institute of Technology, Bombay

Experiment 1: Power measurement in balanced 3-phase circuits

Post Lab Activity

Activity 1:

Design and simulate a circuit in LT Spice to measure active power, apparent power and power factor of a single phase RL load. The power should be calculated using sensed voltage and current values and appropriate analog signal processing circuits (Op-amps preferably) must be used to realize:

- Voltage and current sensing
- Analog multiplication of V and I (to get instantaneous power)
- Averaging circuit (to extract active power)
- Peak detector circuits (to obtain V_{rms} and I_{rms} via peak values)
- Calculation of apparent power and power factor ($pf = P/S$)

Design the circuit to operate correctly for a wide voltage range (80 V to 250 V rms) under varying load conditions.

Activity 2:

Connect a capacitor in parallel to the RL load. Choose a value of capacitor for the circuit to operate at unity power factor (upf). Validate the upf operation in the simulation.

Value of load resistor = Given in the attached PDF

Value of load inductor = 1 H

Note the following observations in simulation:

1. Variation of active power on addition of capacitor
2. Variation of reactive power on addition of capacitor

Submission:

Please submit a report (in pdf format) on how you tested your simulation along with the relevant waveform. Submit your simulation as an electronic file .asc. We will view, test and assess your shared file, but we will not debug it.

Sr. no.	Roll	Name	Value of Resistor (In Ohms)
1	20D070080	Sonu Meher	620
2	24B1264	Joshi Ojas Sandeep	42
3	25B1201	Shaik Sana Taqhia Tahliil	510
4	25B1202	Cliahta Nagi Reddy	62
5	25B1203	Ansh Mankar	420
6	25B1204	Gunjesh Tamle	42
7	25B1205	Dev Tinker	1500
8	25B1206	Bedika Bhatia	75
9	25B1207	Atharva Airen	120
10	25B1208	Yash Rajesh Shirishkar	15
11	25B1209	Nagamwar Haritej	120
12	25B1210	Latisha Meena	62
13	25B1211	Ritanya Danasekaran	260
14	25B1212	Telagathoti Vikram Levi	112
15	25B1213	Jay Nilesh Wakchaure	680
16	25B1214	Kotha Shishir	100
17	25B1215	Kynita K	120
18	25B1216	Nerkar Om Pravin	15
19	25B1217	Kartikay Raina	1500
20	25B1218	Pinakin Sharma	100
21	25B1219	Aman Pratap Singh	680
22	25B1220	Suhas Kashyap	10
23	25B1221	Pranav Poojari	260
24	25B1222	Saket Vempalli	22
25	25B1223	Kuruva Jeevan Kumar	260
26	25B1224	Aditya Avinash Sonule	90
27	25B1225	Gitte Prashant Parmeshwar	420
28	25B1226	Nidhi Ashok Lohana	10
29	25B1227	Anagha S Karnis	220
30	25B1228	Sailasri P	75
31	25B1229	Bhavya Kataria	330
32	25B1230	Gurjot Singh Dhillon	100
33	25B1231	Aaryan Sandeep Bhattad	260
34	25B1232	Yatharth Kumawat	47
35	25B1233	Ahaan Singh	470
36	25B1234	Pushkal Ladha	62
37	25B1235	Devendra Sukhdeo Deshmukh	380
38	25B1236	Gandhi Dhruv Dinesh	112
39	25B1237	Siddharth Bharadwaj	380
40	25B1238	Vartak Gandhar Vivek	75
41	25B1239	Simran Suhani	260
42	25B1240	Priyanshu Latta	15
43	25B1241	Sourav Kumar Bairwa	420
44	25B1242	Domada Harsha Vardhan	47

Sr. no.	Roll	Name	Value of Resistor (In Ohms)
45	25B1243	Sauransh Alok Goenka	1000
46	25B1244	Dhruva Prakash	47
47	25B1245	Ishaan Tulara	680
48	25B1246	Anuj Shivprasad Pagar	62
49	25B1247	Aryan Dharmesh Anarkat	330
50	25B1248	Pranjal Vishnuprasad Sikchi	10
51	25B1249	Pranav Mintri	680
52	25B1250	Sangrulkar Advay Amit	68
53	25B1251	Srinath Sridhar	1500
54	25B1252	Natu Dhruv Amol	56
55	25B1253	Trijal Pravinkumar Bhagat	560
56	25B1254	Hrishikesh L	100
57	25B1255	Shamayita Ray	330
58	25B1256	Lakshya	62
59	25B1257	Amjad Ali	1000
60	25B1258	Sushant Sudhakar Holkar	42
61	25B1259	Bagul Pranav Kiran	470
62	25B1260	Ritvik Nainawatee	10
63	25B1261	Konjarla Shankar Manik	220
64	25B1262	Shreyas Kansal	47
65	25B1263	Soniya Regar	420
66	25B1264	Rishabh Meena	47
67	25B1265	Goggala Hemanth	560
68	25B1266	Rachit Kumar	90
69	25B1267	Varanasi Lakshmi Narayana Nikhilesh	330
70	25B1268	Shagun Nilesh Tapadia	22
71	25B1269	Aayush Ravi Chaudhari	220
72	25B1270	Nagpurkar Vidit Vikram	100
73	25B1271	Kabir More	1500
74	25B1272	G Saikiran	82
75	25B1273	Kavyam Agrawal	620
76	25B1274	Sreedevi S	68
77	25B1275	Manish Khileri	680
78	25B1276	Harshil Agrawal	33
79	25B1277	Pankaj Das	1000
80	25B1278	Lokesh N S	10
81	25B1279	Sohan Kalidas Chelekar	220
82	25B1280	Manu	75
83	25B1282	Mantra Nimesh Shah	120
84	25B1283	Poli Sai Mahanth Reddy	22
85	25B1284	Harmish Kamalbhai Majithiya	220
86	25B1285	Narige Raj Kumar	62
87	25B1286	Gangidi Prathyush Reddy	220
88	25B1287	Bhukya Charitha Sree	100
89	25B1288	Harsh Jha	420

Sr. no.	Roll	Name	Value of Resistor (In Ohms)
90	25B1289	Nishtha Bansal	42
91	25B1290	Guddanti Chetan Naga Anirudh	380
92	25B1291	Aayush Kumar	22
93	25B1292	Chirantan B K	560
94	25B1293	Aman	75
95	25B1294	Soumy Gupta	1000
96	25B1295	Brajaraj Pal	100
97	25B1296	Ruchit Kateshiya	330
98	25B1297	Vedant Shrikant Bhat	33
99	25B1298	Ashwini Kumar	380
100	25B1299	Riddhimaa Mishra	112
101	25B1300	Challa Ansika	330
102	25B1301	Laxmi Sudeeksha Kanduri	15
103	25B1302	Gavali Samruddhi Ganesh	150
104	25B1303	Parth Chetan Chanape	15
105	25B1304	Ayush Agrawal	120
106	25B1305	Vedant Dhirde	68
107	25B1306	Aaditya	260
108	25B1307	Sandesh Shete	33
109	25B1308	Sarthak Goel	1500
110	25B1309	Yash Kothari	62
111	25B1310	Deependra Kumar Vastrakar	150
112	25B1311	Nishant Miglani	42
113	25B1312	Chetaansi Mina	260
114	25B1313	Rajveer Singh Choudhary	62
115	25B3901	Ayush Pati	1000
116	25B3902	Rishabh	62
117	25B3903	Tanishka Mittal	420
118	25B3904	B S Dhanush	10
119	25B3905	Vaibhav Ramdas Patil	150
120	25B3906	Vivaan Bhatia	75
121	25B3907	Santhosh Senthilmurugan	120
122	25B3908	Patankar Siddhant Dheeraj	75
123	25B3909	Himaghn Gupta	470
124	25B3910	Siddhi Mahesh Hivarekar	22
125	25B3911	Jahanvi Gahlout	560
126	25B3912	Harsh Pravin Mhaske	82
127	25B3913	Srishti Yadav	420
128	25B3914	Abhishek Yadav	68
129	25B3915	Tirth Siddharth Sheth	1000
130	25B3916	Gaurish Bansal	75
131	25B3917	Uddipta Talukdar	510
132	25B3918	Raj Gaurav Singh	62
133	25B3919	Atulya Jidnyasu Kamble	220
134	25B3920	Srushti Sanjay Jadhav	22

Sr. no.	Roll	Name	Value of Resistor (In Ohms)
135	25B3921	Prince Silolia	510
136	25B3922	Manas Mahesh Jagtap	18
137	25B3923	Sashreek Sikder	680
138	25B3924	Vansh Goel	33
139	25B3925	Rachel Singhal	470
140	25B3926	Parakram Jangir	10
141	25B3927	Adarsh Gautam Jha	510
142	25B3928	Arjun Gupta	100
143	25B3929	Sadashiv Nagar	330
144	25B3930	Shivoham Shinde	18
145	25B3931	Sapavath Pavan	1000
146	25B3932	Md Imbisat Azim	90
147	25B3933	Ayush Dawar	220
148	25B3934	Akshat Jain	33
149	25B3935	Udani Aarush Rajenkumar	510
150	25B3936	Nitin Bamnawath	22
151	25B3937	Rajat Tiwari	380
152	25B3938	Prayansh Sagar Singh	112
153	25B3939	Binayak Sahoo	260
154	25B3940	Pathalavath Jaswanth Naik	75
155	25B3941	Harshita Sinha	150
156	25B3942	Hari Kishan Baiju	112
157	25B3943	Shreyash Paliwal	1000
158	25B3944	Rajat Bhorla	112
159	25B3945	Ishwar Kumar	620
160	25B3946	Polu Anjali	82
161	25B3947	Prasoon Singh	1000
162	25B3948	Azim Ahmed	62
163	25B3949	Abhineet Bhausaheb Avhad	1500
164	25B3950	Mehta Sneha Hardik	112
165	25B3951	Ingavale Vasundhara Dilip	560
166	25B3952	Venkat Suryaansh Thota	47
167	25B3953	Gore Paras Milind	380
168	25B3954	Tathagata Roy	22
169	25B3955	Soumay Raniwal	220
170	25B3956	Dhairya Chauhan	22
171	25B3957	Racheti Cheriyan	260
172	25B3958	Eklavya Klair	47
173	25B3959	Pradyun Divakara Shetty	330
174	25B3960	Avi Agarwal	112
175	25B3961	Pranav Pavan Jaju	380
176	25B3962	Pranshu Das	82
177	25B3963	Saksham Kailash Tambe	470
178	25B3964	Jiya Lohote	90
179	25B3965	Arnav Jagtap	330

Sr. no.	Roll	Name	Value of Resistor (In Ohms)
180	25B3966	Navinya Devanand Desai	47
181	25B3967	Tanav Gupta	260
182	25B3968	Iniyan V T	15
183	25B3969	Keshav Jindal	380
184	25B3970	Nikhil Dasari	33
185	25B3971	Mugdha Bansal	380
186	25B3972	Sanskriti Yuvraj Shinde	90
187	25B3973	Ratnala Srisiva Yashasri	510
188	25B3974	Suhan Deep Singh Sahni	68
189	25B3975	Lad Atharva Dinesh	120
190	25B3976	Reyansh Mittal	18
191	25B3977	Malvia Arjav Himatbhai	120
192	25B3978	Khushi Pankaj Jadhav	82
193	25B3979	Aditya Jha	150
194	25B3980	Tushar Dhingra	82
195	25B3981	Santhosh T	1500
196	25B3982	Ingle Sai Santosh	56
197	25B3983	Leena Dahire	420
198	25B3984	Krishna Kumar Jangid	90
199	25B3985	Sridhu Nethra D	560
200	25B3986	Medhansh	33
201	25B3987	Anay Sajith S	220
202	25B3988	Ashutosh Sharma	90
203	25B3989	Harith Syam M	510
204	25B3990	Siddhartha Satish Nair	15